ANANDA SANGLI

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SUMMARY

Robotics engineer with experience developing ROS 2 software, integrating electromechanical systems, and implementing perception and localization pipelines for autonomous platforms.

EDUCATION

Northeastern University — Boston, MA
Master of Science in Robotics — Expected: Dec 2026

University of Illinois at Urbana-Champaign — Champaign, IL
Bachelor of Science in Mechanical Engineering — May 2023

SKILLS

Software: C++, Python, ROS 2, Linux, Git, C, CAN, Simulink
Robotics: Perception, Localization, Sensor Fusion, State Estimation, Motion Planning, Controls, System Integration
Computer Vision & ML: OpenCV, YOLOv8, PyTorch, Gaussian Process Regression, Kalman Filtering
***Hardware & Prototyping*:** dSPACE, TI C2000, Raspberry Pi, Arduino, SolidWorks, Fusion 360, 3D Printing, Laser Cutting, CNC

EXPERIENCE

TOYOTA RESEARCH INSTITUTE – Platform Research, Human-Interactive Driving — Los Altos, CA
System Integration Engineer | Portfolio — Sep 2024 – May 2025
Intern | Portfolio — Dec 2023 – Aug 2024

- Integrated electrical, mechanical, and software subsystems for prototype autonomous battery electric vehicles, supporting system bring-up, hardware validation, and debugging
- Implemented brake-by-wire architecture using Simulink and CAN, integrating pedal position sensing with low-level controllers and brake ECUs to control caliper pressure
- Developed a ROS 2 Python package for a pressure-sensitive load-cell handbrake using publisher/subscriber nodes, multithreading, and thread-safe queues
- Investigated autonomous-driving perception & localization pipelines including MPC and SAM 2 segmentation for lane boundary labeling
- Designed and fabricated vehicle computer enclosure using SolidWorks, laser cutting, and mechanical fabrication, improving wiring organization and serviceability
- Integrated an additional 35.7kWh 400V Li-ion battery system, collaborating across mechanical and electrical teams

HULLBOT — San Francisco, CA
Underwater Robot Operator | Portfolio — Feb 2025 – May 2025

- Teleoperate underwater robots with depth, cruise, and hull-hold control capabilities to clean the San Francisco Bay ferries by promoting gentle & early removal of biofouling, reducing drag and fleet emissions by 10-40%

IMPOSSIBLE OBJECTS — Remote
Application Engineer | Portfolio — Dec 2023 – Apr 2024

- Utilized Fusion 360 to design 3D-printed wave solder PCB pallets for customers, reducing setup time, bridging, board warping, improving solder flow, and protecting heat-sensitive equipment

PROJECTS

Maritime Robotic Perception – YOLOv8 + IMU Horizon Stabilization | Portfolio | GITHUB — May 2026 – Aug 2026

- Built a C++ maritime perception pipeline combining IMU-guided horizon stabilization, YOLOv8 detection, and Kalman filtering to detect and track boats, buoys, docks, and other navigation hazards under camera motion
- Evaluated stabilization and temporal filtering approaches for robust obstacle detection from a moving maritime platform

Wearable Terrain Recognition – IMU + TCN | Portfolio | COLAB — Jan 2026 – Apr 2026

- Built a wearable terrain-recognition system using four IMUs and a Raspberry Pi 4, implementing C++ data acquisition at 250 Hz and classifying four terrain types from gait dynamics

- Compared 1D CNN, LSTM, and TCN models, achieving 96% accuracy with a Temporal Convolutional Network for IMU-based terrain classification

Wi-Fi Indoor Localization – ROS 2 + Gaussian Process Regression | Portfolio | GITHUB Sep 2025 – Dec 2025

- Developed a learning-based indoor localization system estimating 2D position from BSSID-indexed RSSI measurements using multi-output Gaussian Process Regression
- Deployed a ROS 2 Prediction node within a sensing and filtering pipeline, incorporating model uncertainty for downstream state estimation

UR10e Camera Orbit Trajectory – Kinematics & Motion Planning | Portfolio | GITHUB Sep 2025 – Dec 2025

- Implemented forward/inverse kinematics for a UR10e using DH parameters & damped least-squares Newton-Raphson IK
- Planned a 400-waypoint 3D inspection trajectory maintaining end-effector orientation toward a target while avoiding singular configurations